PG No. 19 C_{3v} $3m$ (3m1 setting) [trigonal]

* character table

C_{3v}	$1(1)$	$3^+_{001}(2)$	$m_{100}(3)$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

* polar $\leftrightarrow$ axial conversion

A_1 (A_2) A_2 (A_1) E (E)

* symmetric product

	A_1	A_2	E
A_1	A_1	A_2	E
A_2		A_1	E
E			A_1+E

* anti-symmetric product

A_1	A_2	E
$-$	$-$	A_2